JOURNAL OF GEOPHYSICAL RESEARCH, VOL. 96, NO. B4, PAGES 6011–6022, APRIL 10, 1991

Paleomagnetic Results From Mesozoic Strata in the Mérida Andes, Venezuela

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We report paleomagnetic results from Jurassic through Cretaceous sedimentary rocks (nine sites) in the Mérida Andes, Venezuela. Detailed thermal and alternating field demagnetization yielded well-grouped directions of both polarities. The mean of the normal directions is statistically identical to the mean of the reversed directions and the site means clusters better after structural correction, both at the 95% significance level, indicating that the primary magnetization has been isolated. The mean pole position lies at 86.5°N, 151.1°E, A_{95} = 5.6°. Combining this pole position with 10 other poles from the southern part of cratonic South America establishes a mean pole for the Late Jurassic through Cretaceous at 87.6°N, 211.0°E, A_{95} = 3.3°. These results imply, within the error limit of the paleomagnetic data, that South America is indeed one rigid plate and that the motion of this plate was restricted to east-west translation since the opening of the South Atlantic Ocean.

INTRODUCTION

The paleomagnetic data base for South America is still rather limited. For the Jurassic and Cretaceous periods there are 15 pole positions from the stable part of the continent which pass the "confidence filter" of *Beck* [1988]. Beck's selection criterion requires that the mean pole is based on three or more independently derived pole positions, that the samples were demagnetized, and that each site mean has a 95% circle of confidence of < 20°. Except for two Early Jurassic determinations, all pole positions were obtained from sampling sites located south of 6°S. This leaves the northern half of South America void of paleomagnetic data which could be used to constrain the motion of the South American plate since the opening of the South Atlantic Ocean (~125 Ma [*Cande et al.*, 1989]). We report here paleomagnetic results from Jurassic and Cretaceous sedimentary rocks from the Mérida Andes in Venezuela which were obtained under a joint program between the Universidad Simón Bolívar, Caracas, and the University of Texas at Austin.

GENERAL GEOLOGY

In northwestern South America, the Andes branch into three distinct mountain ranges, the western, central, and eastern Cordilleras of Colombia. The northeastern part of the eastern Cordillera is known as the Venezuelan or Mérida Andes. The central portion of the Mérida Andes consists of Precambrian crystalline rocks, the Mérida Arch, which is thought to be an extension of the Guayana shield [*Zambrano et al.*, 1972]. The Mérida Arch is overlain in the northeast and the southwest by sedimentary and metamorphic rocks of Paleozoic, Mesozoic, and Paleogene age (Figure 1). The Cretaceous strata onlap onto the Mérida Arch indicating that it was a positive structural feature during this time. Except for Pleistocene glacial moraines and fluvial terraces, rocks of post-Eocene age are virtually absent [*Bellizzia et al.*, 1976, *Kohn et al.*, 1984].

The oldest Mesozoic rocks are the red beds of the Jurassic/Early Cretaceous La Quinta Formation [*Gonzáles de Juana et al.*, 1980; *Maze*, 1984] which rest unconformably on Paleozoic or older rocks (Figure 2). The predominantly clastic Río Negro Formation of Neocomian through Barremian age is in unconformable contact with the La Quinta Formation. The Aptian Apón Formation (limestones and shales) indicates a marine transgression. During the Late Aptian and much of the Albian detritus from the Guayana shield dominates the sedimentary sequence in the Andes (Aguardiente Formation). Toward the north and west the strata become progressively more calcareous. The Late Albian is represented by the shallow marine limestones of the Maraca Formation. Most of the northern and northwestern rim of South America was covered in Cenomanian through Santonian time by the marine limestones of the La Luna Formation and its equivalents. The La Luna Formation is one of the source rocks for the oil in Venezuela and it is well known for its rich ammonite fauna [e.g., *Renz*, 1959, 1982]. Toward the top of the La Luna in the southern part of the Andes are localized siliceous beds, often phosphatic; one such sequence is the Quevedo Formation. The Cretaceous sequence closes with the open marine shales of the Colon Formation. The Jurassic/Cretaceous stratigraphy is summarized in Figure 2 (based on the work by *Gonzáles de Juana et al.*, [1980]) which also shows the geomagnetic reversal sequence [*Harland et al.*, 1982].

PREVIOUS RESULTS FROM THE MERIDA ANDES

The first paleomagnetic data from Venezuela were reported by *Creer* [1962] who measured a suite of samples from the La Quinta Formation. The samples were not demagnetized. *Hargraves and Shagam* [1969] collected La Quinta samples at four localities in the southern Mérida Andes. The authors argue for remagnetization in the Tertiary, an interpretation that is difficult to justify in view of the large differences in

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Paper number 90JB02282.
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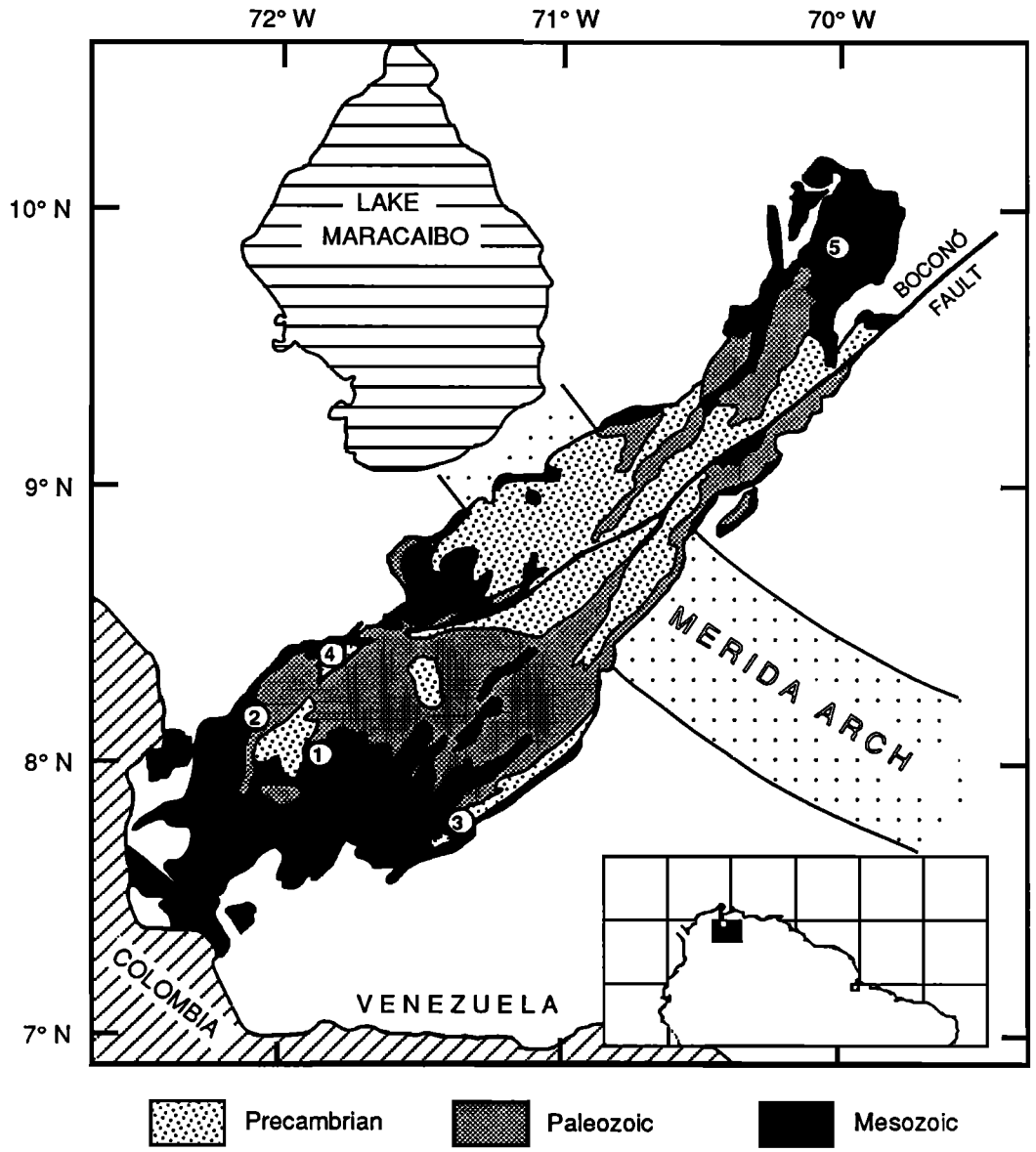


Fig. 1. Generalized geological map of the Mérida Andes after *Bellizzia et al.* [1976] and *Gonzáles de Juana et al.* [1980] and location of sampling areas. Area 1 is between La Grita and Pregonero; area 2 near La Quinta and La Fría; area 3 along Río Caparo upstream of Santa María de Caparo near La Vueltosa; area 4 near Zea; area 5 is near Barbacoas. Insert shows northern part of South America.

the mean directions of the sites. In the same area *Shagam and Hargraves* [1970] sampled Late Paleozoic red beds, the Mérida facies and the Sabaneta facies. The results from the Mérida facies have to be discarded on statistical grounds ($N_{samples} = 5$, $\alpha_{95} = 33°$). The Sabaneta data are interpreted to be the result of Tertiary remagnetization and the authors cautiously suggest that the results may indicate a clockwise rotation for the Mérida Andes. Additional results from the La Quinta Formation were presented by *Creer* [1970], but he does not consider the data reliable.

EXPERIMENTAL PROCEDURES

For our paleomagnetic study, we collected Mesozoic strata at 18 sites for a total of 306 samples (see Figure 1 for location of sampling areas). Half of the samples were measured at the Universidad Simón Bolívar using a Molspin magnetometer and a Schonstedt thermal or alternating field demagnetizer located inside three sets of Helmholtz coils. At the University of Texas, an ScT cryogenic magnetometer and Schonstedt furnace or AF unit were used, all of which are housed inside a magnetically shielded room. All samples were stepwise demagnetized, and the data were analyzed by the principal component method. Samples were excluded from the analysis of the site mean if either no stable component of magnetization could be identified or if their direction differed from the site mean by more than two angular standard deviations.

MAGNETIC MINERALOGY

The magnetic mineralogy of a limited number of samples was determined by analyzing their coercivity spectra and by microscopic examination of thin sections. Following the

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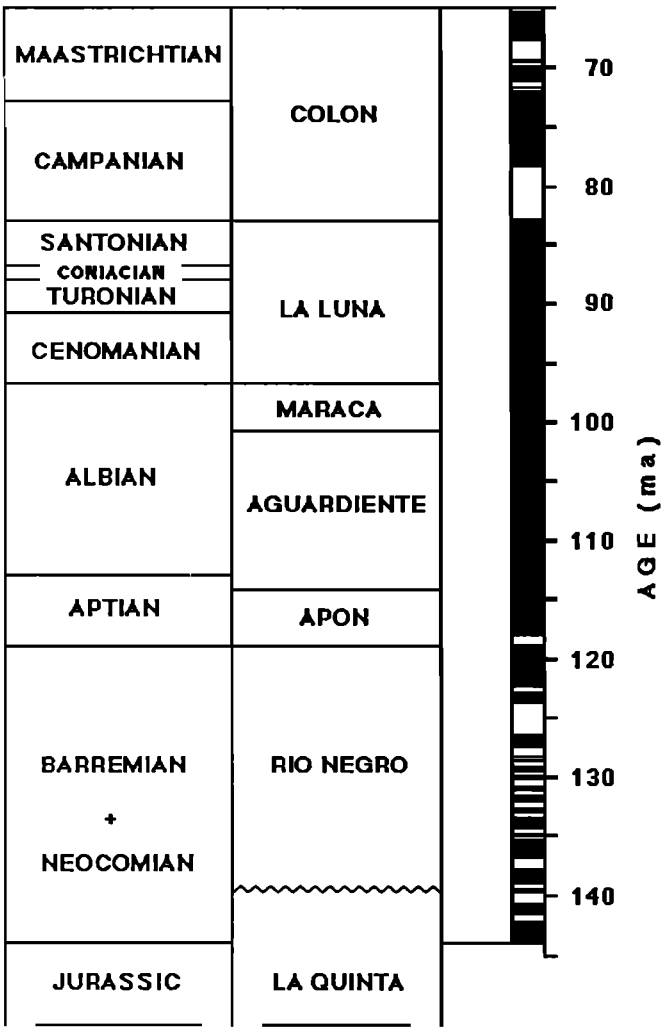


Fig. 2. Jurassic/Cretaceous stratigraphic column for the Mérida Andes [*Gonzáles de Juana et al.*, 1980] and the Cretaceous geomagnetic reversal time scale [*Harland et al.*, 1982].

procedures developed by *Dunlop* [1972], the samples were given an isothermal remanent magnetization (IRM) in fields up to 1 T. The coercivity spectrum is approximated by the incremental magnetization in 0.1- or 0.05-T intervals. Based on the results from synthetic samples of well-defined grain size, *Dunlop* [1972] divides the coercivity spectrum into three ranges, the magnetically soft ($0 < H_{cr} < 0.1$T), intermediate ($0.1 < H_{cr} < 0.3$T), and hard ($H_{cr} > 0.3$T) fraction, corresponding approximately to the contribution of magnetite or maghemite, coarse-grained hematite [specularite], and fine-grained hematite (grain size < 5 μm). In interpreting the spectra it is important to remember that the saturation magnetization of magnetite and maghemite is over 100 times that of hematite.

Figure 3 shows the IRM acquisition curve for two samples from the La Quinta Formation and their coercivity spectra. Sample LQP2-12 is dominated by specularite and also exhibits a significant contribution to the remanence by finer-grained hematite. The spectrum of a sample from Río Caparo (LQC1-15) is dominated by low-coercivity grains, but the presence of high-coercivity grains is clearly apparent. There may be a small contribution of magnetite to the low coercivity, but none of the samples from this site showed an intensity decrease during thermal demagnetization upon crossing the magnetite Curie point. Thus the remanent magnetization of these red bed samples is carried by hematite. In contrast, samples from the Quevedo Formation (QV1-01) are dominated by magnetite.

The La Luna samples from the Barbacoas area (locality 5) are mainly wackestones with allochems of foraminifera, mollusc fragments and echinoderm fragments (in decreasing order of abundance). Hematite typically occurs as a replacement of pyrite framboids which are generally found inside the allochems; it is also observed along stylolites. Most pyrite formed early in the marine diagenesis. Bleached zones adjacent to calcite veins and their cross cutting of the stylolites implies that the hematite formed before extensive chemical compaction and prior to fracturing. In some thin sections, small amounts of a later generation of hematite were also observed.

The coercivity spectrum analysis confirms the petrographic observations. The spectra clearly point to the predominance of hematite, both coarse-grained and fine-grained (Figure 3, LB01-03 and LB01-17). Sample LL08-08 may contain traces of magnetite, but again, no evidence for magnetite was found during thermal demagnetization.

PALEOMAGNETIC RESULTS

La Quinta Formation

The La Quinta Formation was sampled at three localities in the southern Mérida Andes. The first three sites are near the type locality (area 1 and 2) and the third site is located some 80 km to the east along Río Caparo (area 3, Figure 1).

The samples from site LQP1 (area 1) cover a strati-graphic thickness of about 70 m. Thermal demagnetization revealed two antiparallel directions of magnetization (Figure 4). The mean of the normally magnetized samples is statistically indistinguishable from the mean of the reversed samples as verified by the F-test. Site LQP2 (area 1) spans 10 m stratigraphically and yielded very similar results. Nine samples are of normal polarity and five samples are reversed, and the two means are antipodal within 10°. The vector component diagrams for samples from both sites show one predominant component of magnetization. Athough the samples are of rather uniform lithology, some have a distributed unblocking temperature spectrum while others retained most of their intensity up to 660°C and even up to 675°C (Figure 4). Compared to the results from the first two sites, the data from the third site (LQF1, area 2) scatter both before and during thermal demagnetization (see Table 1).

Above the flood plain of Río Caparo, an isolated outcrop of La Quinta strata (LQC1, area 3) exposes 20 m of section. The in situ NRM (natural remanent magnetization) directions are strongly biased towards the present geomagnetic field direction. Upon thermal demagnetization, two statistically identical antipodal directions emerge (Figure 5). It is important to note that the directions of magnetization before tilt correction are clearly not the present field direction, an indication that the samples have not been remagnetized since their deformation.

Río Negro Formation

Near the LQC1 site we collected 12 samples in the lower part of the Río Negro Formation (site RN1, area 3). Eight samples define a well-grouped reversed direction of magnetization (Figure 5). The second sampling locality is along the road near Zea on the western slope of the Andes (area 4, site RN2). Fourteen out of 24 samples did not yield a characteristic direction. The 10 remaining samples are all

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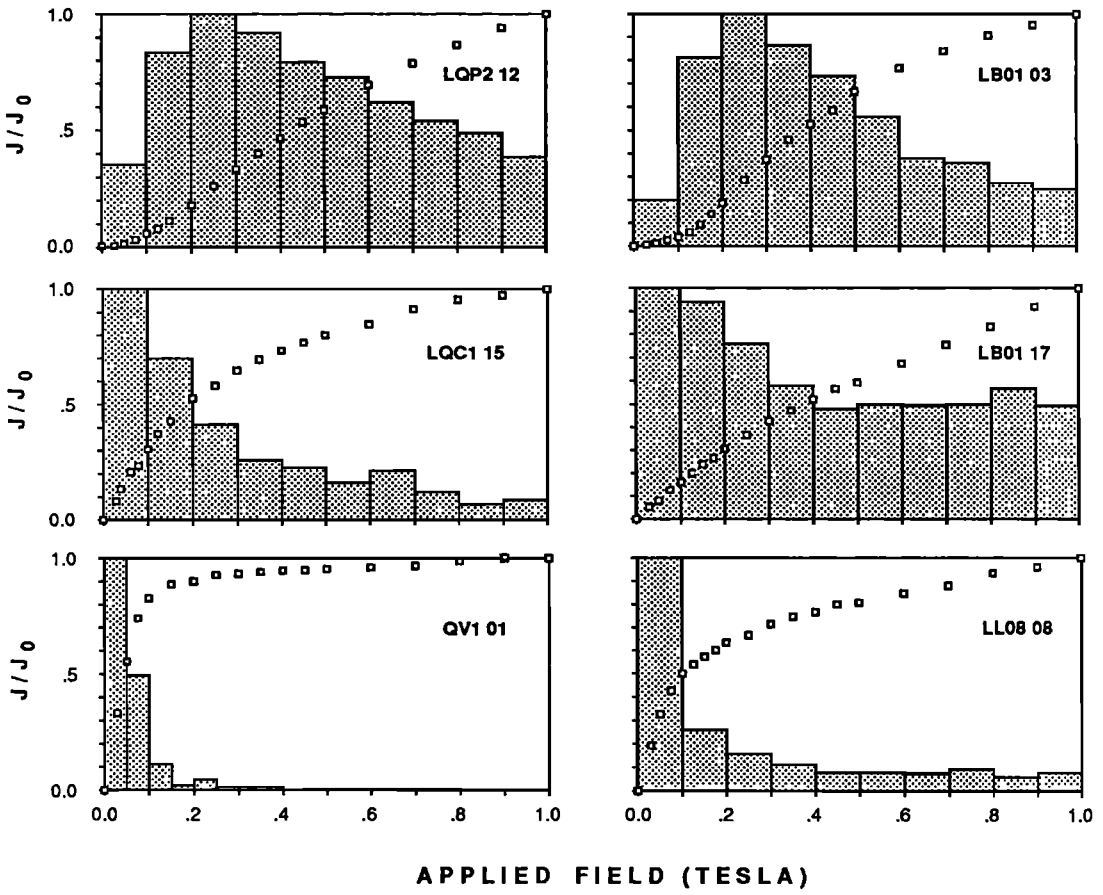


Fig. 3. Normalized isothermal remanent magnetization (IRM) acquisition curves (squares) and coercivity spectra (histograms) for two La Quinta samples (LQP2-12, LQC1-15), a sample from the Quevedo Formation (QV1-01), and three La Luna samples (LB01-03, LB01-17, LL08-08).

normally magnetized. Before structural correction, their directions are closely aligned with the present geomagnetic field direction. It seems clear that the samples from this weathered outcrop have been recently remagnetized (compare La Luna site LL06).

Aguardiente Formation

At the first locality (area 1, site AGP1) we collected 13 samples. Their directions of magnetization fall into two antipodal groups. Figure 6 shows two examples of vector component diagrams. Compared to the data from all other outcrops, these data cluster poorly (see Table 1).

Some 30 km away, we collected another 48 samples at a long road cut (area 2). None of the samples from this strongly weathered outcrop is of clear reversed polarity in contrast to site AGP1. Before structural correction, one third of the samples are centered around the present field direction and the remaining directions scatter resulting in a value of k=2.3 for Fisher's precision parameter. It is quite obvious that this site has been remagnetized and it will be deleted from further consideration. A third site near Barbacoas in the northern Andes (area 5) yielded identifiable characteristic directions in only 5 of 10 samples and the distribution of directions is random (k=1.2).

La Luna Formation

The first three outcrops of the La Luna Formation (site LL07, LB01, LB02]) were collected from a section exposed near Barbacoas (area 5). Before structural correction, the NRM directions from site LL07 cluster near the present field directions. Upon thermal demagetization, 6 of the 14 samples move toward a reversed direction. Starting at 400°C these directions move back to normal and after demagnetization to 600°C the directions form a good cluster. Figure 7 shows the change of the direction of magnetization during progressive demagnetization for one of these samples (LL07-5). All samples from site LB01 are of reversed polarity (Figure 7 and 8*a*). A small present field overprint was removed by 200°C. Site LB02 is of mixed polarity (Figure 7 and 8*b*).

Some 10 km to the west we collected another 13 samples (site LL08). The samples lost between 60% and 90% of their intensity upon heating to only 200°C. After heating to 375°C, the samples were subjected to stepwise alternating field demagnetization to 30 mT. Two samples are of normal polarity, the remainder of reversed polarity (Figure 8*c*). Vector component diagrams of a normal and reverse sample from this site are shown in Figure 7.

La Luna site LL06 is a poorly exposed, severely weathered outcrop near Zea (area 4). Only 8 out of 21 samples yielded a characteristic direction of magnetization. As was the case with the adjoining outcrop of the Rio Negro Formation, the structurally uncorrected directions group around the present field direction and remagnetization seems indicated.

Quevedo Formation

The Quevedo Formation is a rather unusual rock type which we collected at a fresh road cut near La Vueltosa at a

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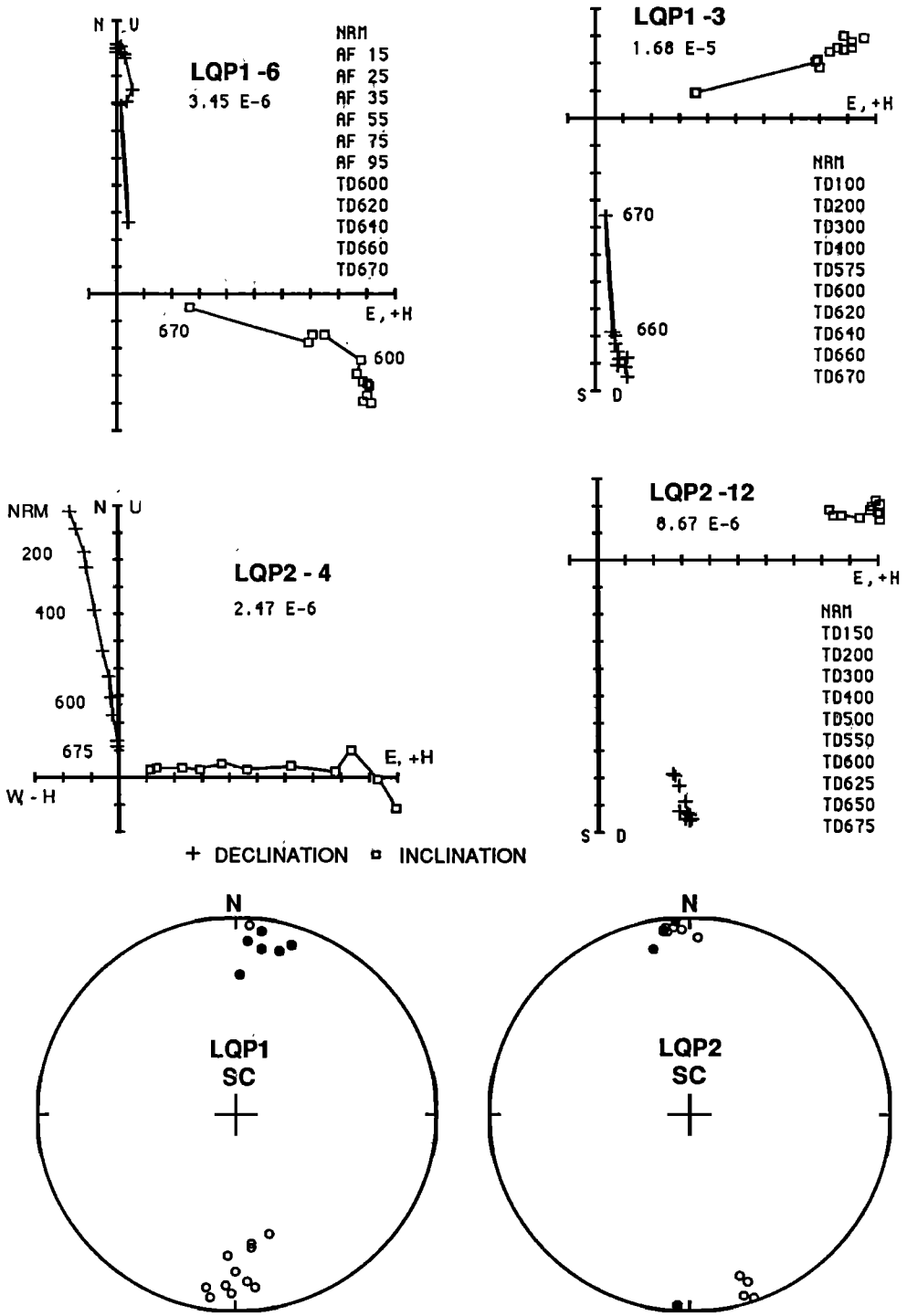


Fig. 4. Orthogonal vector projections for samples from La Quinta sites LQP1 and LQP2. Numbers indicate alternating field demagnetization in millitesla and thermal demagnetization temperatures in degrees Celsius. Number next to sample label is the NRM intensity in Am^2/kg. Bottom of figure shows Schmidt equal area projection of directions of characteristic magnetization for samples from both sites after structural correction. Solid circles are in the lower hemisphere, open circles in the upper. Same conventions apply to all similar figures.

water dam construction site of Río Caparo (area 3, site QV1). The lower portion of the outcrop is a white, microporous argillaceous chert. Cryptocrystalline quartz makes up about 90% of the sample and is the only mineral that was detected in bulk X-ray diffraction patterns. Clay (~10%) and silt (~1%) make up the balance. The rock contains about 1% megapores produced by the dissolution of grains approximately 0.1 mm in diameter. Some pores are rhombic, suggesting the dissolution of carbonate grains, and others are circular or elliptical, resembling the outlines of microfossils (radiolaria, calcispheres, foraminifera). This unit is conformably overlain by a phosphatized sandstone which contains 70% sand-sized grains, 20% microcrystalline apatite cement, and 10% pores, most of which formed by

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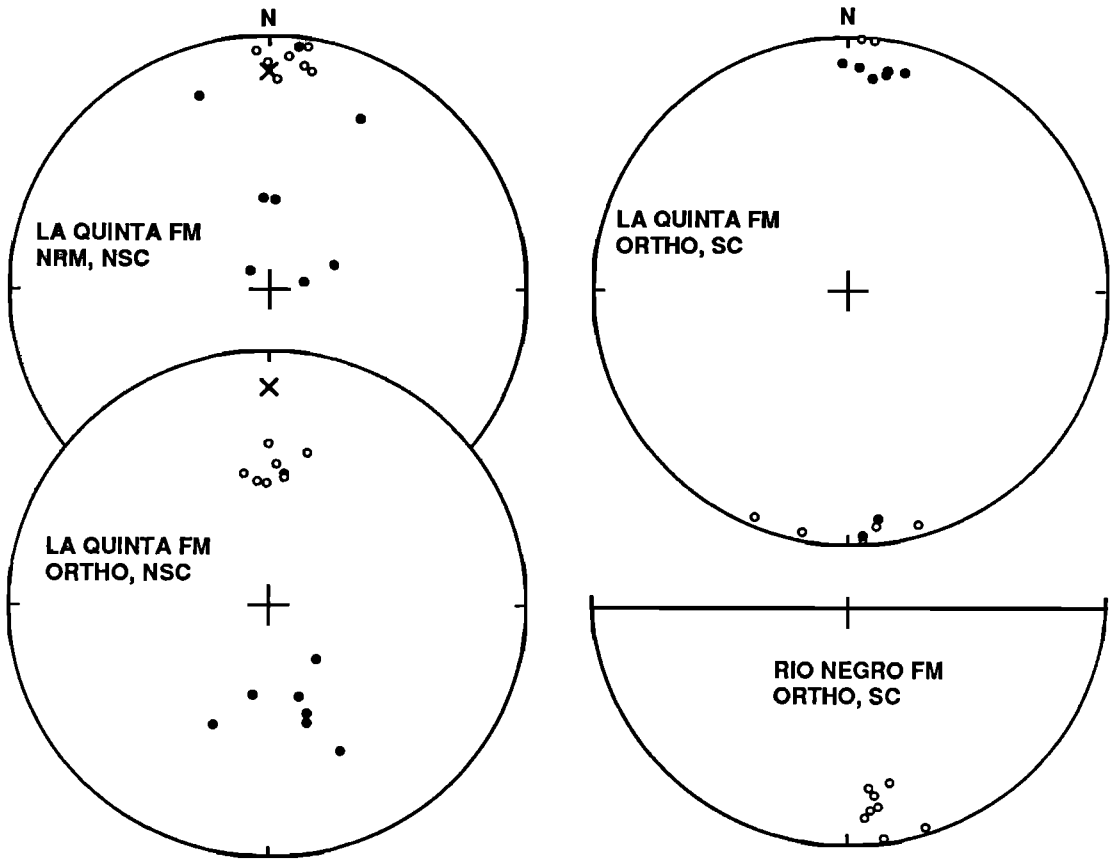


Fig. 5. Directions of magnetization for La Quinta site LQC1 and Río Negro site RN1. NSC means no structural correction, SC means structural correction applied, ORTHO means directions were calculated from orthogonal vector diagrams. Cross represents the dipole field direction.

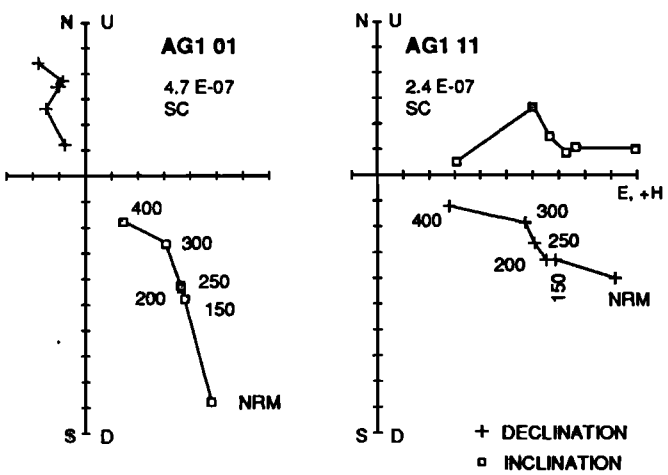


Fig. 6. Vector component diagram for two samples from the Aguardiente Formation.

postdepositional dissolution. The fine-grained sand mode (~50% of rock) is composed nearly entirely of quartz, and the coarse-grained mode (20% of rock) is composed of subequal amounts of phosphatic fossil fragments and structureless, well-rounded, spherical phosphate pellets. Most fossils are phosphatized carbonate grains, but there is a significant amount of vertebrate fragments shown by the presence of haversian canals within the particles.

We collected 11 samples over a 4-m stratigraphic interval in the argiliceous chert beds. All samples are very weakly magnetized (1 - 6 x 10^{-8} Am^2/kg). Magnetic over-printing due to weathering is minimal and thermal demagnetization to 350°C yielded well-grouped directions (Figure 9).

VALIDITY OF PALEOMAGNETIC DATA

The paleomagnetic results with the pertinent statistics are summarized in Table 1. By and large, the data are of very good quality as expressed by large values of Fisher's precision parameter k and small circles of confidence (α_{95}). As we argued earlier, the results from the adjoining sites RN2 and LL06 should be discarded because the samples are remagnetized. The results from sites AGP1 and LQF1 are of much poorer statistical quality than all other results. Because we want the best possible reference pole for the sampling area, we chose to delete them as well because their circles of confidence (α_{95}) are larger than 20° and their k values are very small. But we want to point out that the deletion of these two sites does not change any of the conclusions in the following discussion.

Since the opening of the South Atlantic Ocean the motion of the South American plate has been in the east-west direction [e.g., *Ladd*, 1976, *Cande et al.*, 1989]. Thus the direction of magnetization of rocks of Cretaceous and younger age will be indistinguishable from the present field

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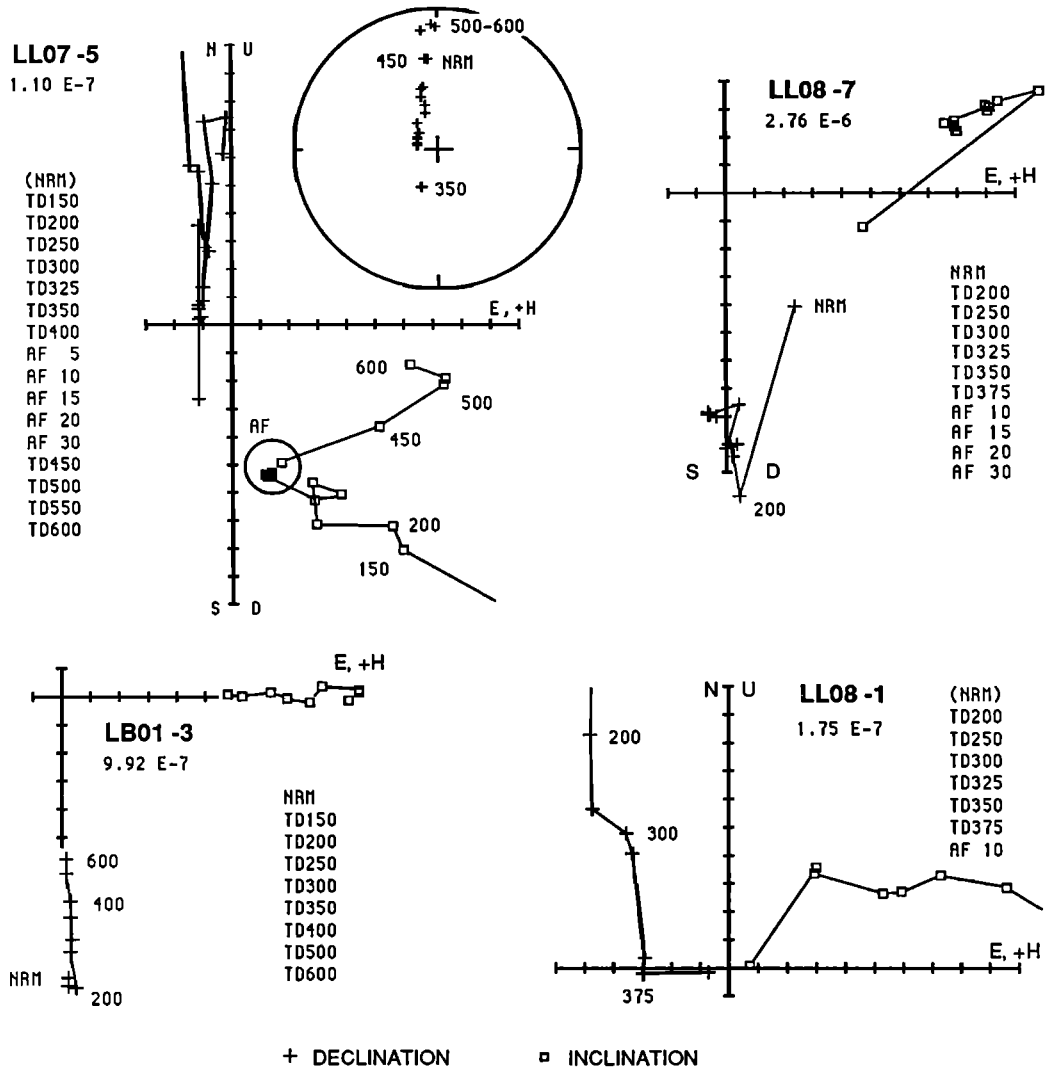


Fig. 7. Orthogonal vector projection for samples from the La Luna Formation. Sample LL07-5 is one of several samples that moved from a normal toward a reversed direction upon low-temperature demagnetization but became clearly normal at higher temperatures.

direction or its inverse. It is therefore important to prove that the observed directions correspond to their geological age as was also pointed out by *MacDonald and Opdyke* [1984]. The nine sites have highly varying bedding attitudes which makes it possible to test whether the magnetization is predeformation or postdeformation. For this analysis all directions have been recalculated for a common site. The precision parameter k before structural correction is k_{before} = 14.5 and after tilt correction k_{after} = 39.1, thus k_{after} / k_{before} = 2.70 which is larger than 2.33, the value for the F-test at the 95% confidence level. The magnetization, therefore, was acquired before folding. The initial phase of deformation of the Mesozoic strata occurred in latest Cretaceous/Early Eocene time [*Shagam*, 1972; *Gonzáles de Juana et al.*, 1980; *Kohn et al.*, 1984] although the main uplift of the Mérida Andes took place only during the last few million years [e.g., *Shagam et al.*, 1984]. Thus the age of magnetization is pre-Early Eocene which strongly argues that the primary magnetization has been isolated.

Another test for magnetic stability as well as an indication that all secondary magnetizations have been removed is obtained by comparing the mean directions of all normally magnetized sites with the mean of all reversely magnetized sites which should be exactly antipodal. In the data description we showed that the sites of mixed polarity indeed consist of two antipodal populations. Applying the same test to the means of the normal and reversed sites or their pole positions verifies that there is no difference between the two populations at the 95% significance level.

THE LA LUNA PROBLEM

The La Luna Formation has been extensively studied because it is one of the source rocks for the oil in the Maracaibo Basin [*Gonzáles de Juana et al.*, 1980]. It is well known for its rich planktonic fauna (ammonites, foraminifera) and the age of the La Luna seems to be well established as ranging from Cenomanian through Santonian [*Renz*, 1959, 1982; *Gonzáles de Juana et al.*, 1980] although *Boesi et al.* [1988] place the top of the La Luna in the upper Campanian. It is also well established that the geomagnetic field was of constant normal polarity from about 118 Ma to 84 Ma, i.e., from basal Aptian to the top of the Santonian [*Harland et al.*, 1982]. Thus it is most surprising that in the

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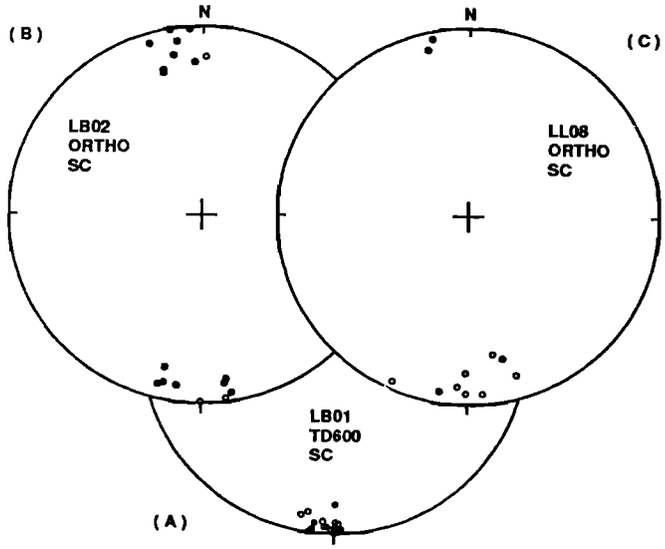


Fig. 8. Structurally corrected directions of magnetization for three La Luna sites.

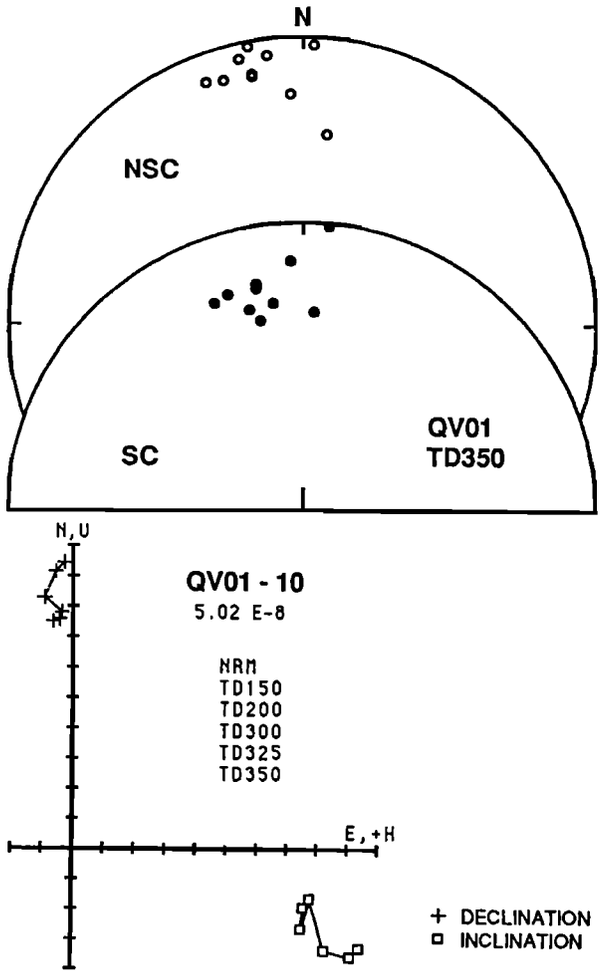


Fig. 9. Directions of magnetization for samples from the Quevedo Formation after thermal demagnetization to 350°C and vector component diagram for sample QV01-10 (note the low intensity of magnetization).

northern Andes as well as in the type area in the Perijá Mountains [*Gose et al.*, 1988; *Perarnau et al.*, manuscript in preparation, 1991] the sampled La Luna strata contain several well-defined magnetic reversals (Figure 10).

Our results can be interpreted in two different ways. In the first interpretation, the remanent magnetization of the La Luna samples was acquired syndepositionally and the age of part of the La Luna Formation is therefore either pre-Aptian or post-Santonian. In this case, the assigned paleontological age would be incorrect. In the second interpretation, the paleontological age is considered correct and the observed magnetization is assumed to have been acquired at some time after deposition.

Although remagnetization of entire sections has been reported from other areas (e.g., Sierra Madre Oriental, Mexico [*Böhnel et al.*, 1990]), it is difficult to visualize a mechanism which will cause remagnetization over an extended period of time such as to yield numerous polarity zones or to selectively remagnetize particular stratigraphic horizons in a section of rather uniform lithology. Also taking into account that the remanence in the samples from the Andes is carried by hematite whereas magnetite is the magnetic carrier in the samples from the Perijá Mountains leads us to prefer the first interpretation.

Because we did not expect any reversals in the La Luna Formation, the sampled interval is not of sufficient stratigraphic extent and detail which might enable us to correlate the reversal sequence with the geomagnetic time scale. The resolution of this conflict is of major importance for Cretaceous biostratigraphy as well as Venezuelan geology. If it turns out that the age of the La Luna Formation is younger (or older) than its generally accepted age then this would require a revision of the Cretaceous paleogeography. If, on the other hand, the rocks were remagnetized at some later but identifiable time, then this may signal the time of major geochemical and geothermal effects related to hydrocarbon maturation and migration. With the current data, we cannot resolve this problem.

DISCUSSION

The sampled strata range in age from Jurassic (La Quinta Formation) to Late Cretaceous (La Luna Formation). The mean pole position of the three La Quinta sites is statistically indistinguishable from the mean pole position of the Cretaceous sites. The equivalence of pole positions implies that no paleomagnetically detectable motion has occurred between the southern and northern Mérida Andes since La Quinta time.

The Mérida Andes are cut by a large number of faults striking subparallel to the long axis of the mountain range. The best known of these faults is the Boconó fault which *Dewey and Pindell* [1985] suggest may represent part of the southern Caribbean plate boundary. In a reply to this paper *Salvador* [1986] pointed out that the known stratigraphy of the Mérida Andes is incompatible with any significant strike-slip offset [see also *Schubert*, 1982]. The mean direction of our data from west of the Boconó fault [the La Luna sites around Barbacoas] and east of the fault (sites LQC1, QV1, RN1) differ by only 2° and are statistically indistinguishable and thus do not support any major strike-slip offset along the Boconó and related faults.

Figure 11 shows the virtual geomagnetic pole positions (VGPs) for the data from this study and also displays the VGPs for all Jurassic and Cretaceous data from stable South America. The South American reference data are summarized in Table 2 and were selected using the criteria of *Beck* [1988]. All radiometric age determinations were obtained by the K-Ar method and have been recalculated according to the

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TABLE 1. Statistical Parameters for Data From the Mérida Andes

Site	Treat	N	R	D	I	k	α_{95}	Latitude	Longitude
La Quinta									
LQP1	ORTHO	7/11/1	17.56	172.8	-15.2	38.7	5.6		
8.1°N, 288.2°E			17.53	182.8	-17.4	36.2	5.8	87.1	1.7
LQP2	ORTHO	9/5/0	13.67	-7.9	16.1	39.1	6.4		
8.1°N,288.2°E			13.75	-8.4	0.1	51.7	5.6	78.4	154.6
LQC1	ORTHO	16/8/0	14.61	-3.8	-49.7	35.6	6.5		
7.8°N, 288.5°E			14.67	3.1	5.9	41.9	6.0	84.3	76.0
LQF1 *	ORTHO	9/7/4	8.74	2.3	55.3	7.1	19.4		
8.2°N, 287.9°E			8.65	-31.3	15.4	6.7	20.2	59.1	199.6
Rio Negro									
RN1	ORTHO	8/0/4	7.91	172.9	-7.9	74.4	6.5		
7.8°N, 288.5°E			7.91	170.4	-20.5	74.3	6.5	80.1	215.6
RN2 *	ORTHO	13/0/3	12.57	-3.0	25.8	28.1	8.0		
8.4°N, 288.2°E			12.57	-2.4	-31.4	28.1	8.0	64.5	113.5
Aguardiente									
AGP1 *	ORTHO	5/6/2	9.21	155.6	8.8	5.6	21.2		
8.1°N, 288.2°E			9.26	152.6	-9.6	5.7	20.9	62.6	193.1
La Luna									
LL07	TD 600	14/0/6	13.72	-3.2	8.8	46.6	5.9		
9.9°N, 299.0E			13.72	6.1	32.3	46.5	5.9	80.3	327.1
LB01	TD 600	20/0/0	19.84	183.0	-14.3	121.3	3.0		
9.9N, 299.0°E			19.84	182.1	-1.1	121.3	3.0	80.5	97.5
LB02	ORTHO	9/9/11	7.30	-2.3	21.5	24.2	7.2		
9.9°N, 299.0°E			17.30	-4.0	0.3	24.2	7.2	79.5	132.3
LL08	ORTHO	2/9/0	10.51	176.4	-1.7	20.5	10.3		
9.9°N, 289.9°E			10.51	176.9	-6.6	20.6	10.3	82.7	135.1
LL06 *	ORTHO	8/0/13	7.62	-1.6	30.4	18.6	13.2		
8.4°N, 288.2°E			7.62	-3.9	-13.6	18.6	13.2	74.2	122.5
Quevedo									
QV1	TD 350	10/0/1	9.78	-9.2	-12.1	41.4	7.6		
7.8°N, 288.5°E			9.78	-9.6	22.7	41.4	7.6	79.7	222.1
Mean		9/4	8.80			40.2	8.2	80.9	139.2
(excludes *)			8.91			85.0	5.6	86.5	151.1

Treat indicates whether the directions were obtained from principal component analysis (ORTHO) are after demagnetization to a particular temperature in degrees Celsius (TD). N is the number of samples of normal polarity/ reversed polarity/rejected from the analysis; R is the resultant vector; D is declination; I is inclination; k is Fisher's precission parameter, α_{95} is the radius of the 95% circle of confidence; Latitude, Longitude are north latitude and east longitude. For each entry, the data in the first line are without structural correction, the data in the seconmd line are with structural correction applied.

tabulation by *Dalrymple* [1979]. The table also includes a recent pole position by *Butler et al.* [1989]. The mean pole position of all South American poles, including the Mérida pole, lies at 86.8°N, 136.1°E with a circle of confidence of A_{95} = 5.9° and a precision parameter k = 42.9. The Early Jurassic poles J1 and J2 lie distinctly east of the other poles which suggests that it may be incorrect to average poles which span the entire Jurassic and Cretaceous periods. Pole K2 seems to be based on incompletely demagnetized data. The 10 virtual geo-magnetic poles determined in this study [*Mendía,* 1978], lie along a great circle spanning 70° of arc and the computation of their mean using Fisher statistics is mathematically incorrect. Deleting these three pole positions from the data set makes pole K6 deviate from the mean by 2.4 angular standad deviations. Using the 11 remaining data points gives a mean pole position at latitude 87.6°N, longitude 211.0°E (A_{95} = 3.3°, k=192.2) which we consider a more reliable pole position for South America for the Late Jurassic through Late Cretaceous.

In his review of South American paleomagnetism, *Beck* [1988] comments on the "streaked" distribution of the Cretaceous poles which *Valencio et al.* [1983] interpreted to reflect genuine apparant polar wander. Realizing that pole K2 is not a valid paleomagnetic pole and that pole K6 can reasonably be rejected because it deviates from the mean of the Cretaceous poles (as well as the Late Jurassic-Late Cretaceous poles]) by more than two angular standard deviations, changes the streaked distribution to a circular one (compare Figure 11*b* and *c*) and the wobbling plate motion of South America, implied by *Valencio et al.* [1983], does no longer seem to be a valid interpretation. The same conclusion was reached by *Beck* [1988].

Our data provide an important new data set because they are the first statistically acceptable results for the northern part of the South American craton (8°N to 10°N) from rocks younger than the opening of the South Atlantic Ocean (~125 Ma [*Cande et al.*, 1989]). All previous sampling areas, corresponding in age to our samples, were located south of latitude 6°S. The agreement of our data from the Mérida Andes with the results from stable South America implies that the Mérida Andes are part of the stable continent. This is not particularly surprising because the Mérida Andes are anchored to the Guayana Shield by the Precambrian Mérida Arch [*Zambrano et al.*, 1972; see Figure 1].

The close agreement among all the data implies that any differential motion between northern and southern South

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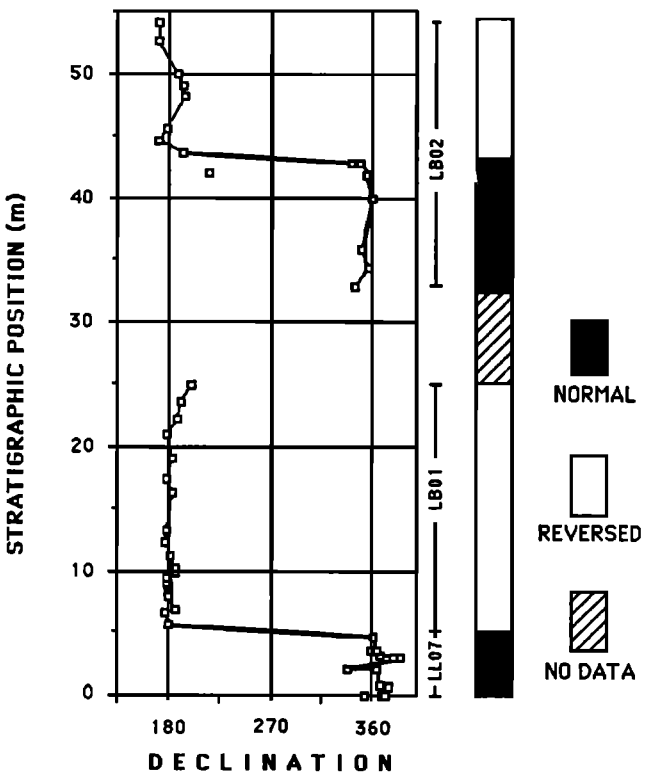


Fig. 10. Magnetic declination (after thermal demagnetization and structural correction) as a function of stratigraphic position for sites LL07, LB01, and LB02 near Barbacoas and inferred magnetic polarity zonation. Ruled pattern indicates an interval with no data.

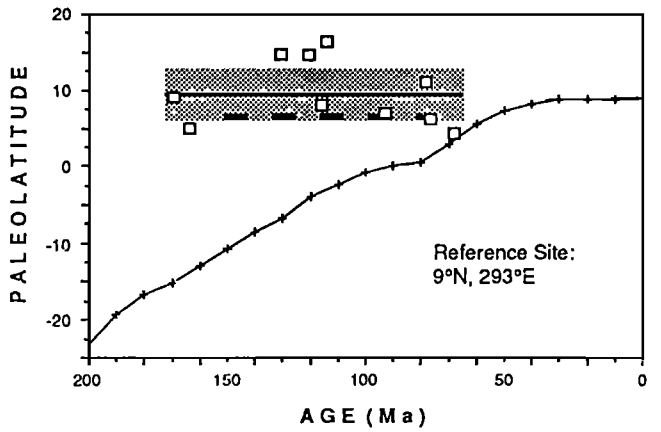


Fig. 12. Comparison of paleoaltitude estimates for South America based on paleomagnetic data (from Table 2) and hotspot tracks (*Morgan*, 1983). Both data sets have been calculated for a reference site at 9°N 293°E (Mérida Andes). The paleomagnetic estimate is shown with an error (grey box) of ±3.3° (95% confidence level). Open squares are the individual data points and heavy dashed line is the mean of the data from the Mérida Andes spanning the age range of the sites.

America, such as might have occurred along the Amazon shearing system [*Grabert*, 1983], is restricted to east-west translations. Within the accuracy of the available data, the Late Jurassic poles cannot be distinguished from the Cretaceous poles. The paleomagnetic pole for the Late Jurassic through Late Cretaceous has a 95% circle of confidence of A_{95} = 3.3° and lies within 2.4° of the geographic pole. The motion of the South American plate since the Late Jurassic is therefore restricted to east-west translation within the error limits of the paleomagnetic data. This conclusion is in contradiction to the absolute plate

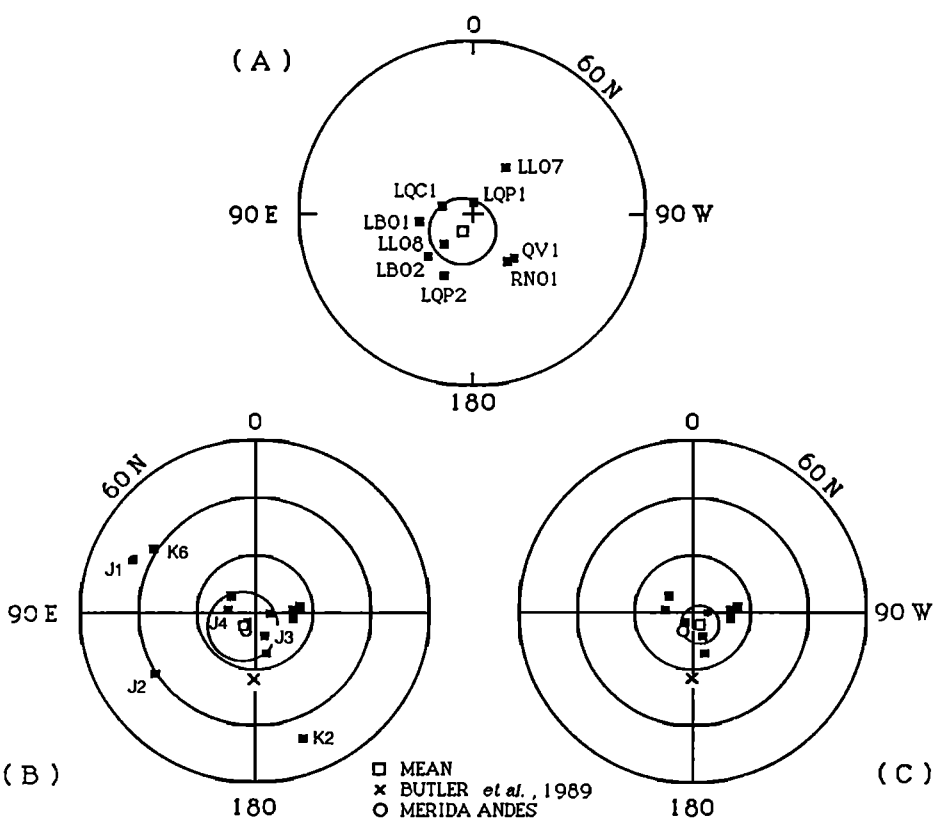


Fig. 11. North polar projection of virtual geomagnetic pole positions (VGPs) for (*A*) the sites from the Mérida Andes, (*B*) all statistically acceptable studies of Jurassic and Cretaceous age from stable South America coded to Table 2 , and (*C*) selected South American sites. The 95% circle of confidence of the mean is shown in all figures. Note that the circles of latitude start at 60°N.

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TABLE 2. Pole Positions of Reference Data From the South American Craton

	Age	Site	N	A_{95}	Latitude	Longitude	Reference
K10	69	45.5°S, 288.7°E	18	14.5	78.7	178.4	*Butler et al.* [1989]
K9	77	21.9°S, 313.4°E	5	10.1	84.8	55.3	*Opdyke and MacDonald* [1973]
K8	79	25.7°S, 394.1°E	6	12.7	87.6	268.7	*Valencio et al.* [1977]
K7	94	8.4°S, 325.0°E	9	4.5	87.6	135.1	*Schult and Guerreiro* [1980]
K6	109	25.7°S, 294.2°E	4	12.0	69.4	57.4	*Valencio et al.* [1977]
K5	<117	32.2°S, 295.9°E	5	7.4	82.4	278.2	*Vilas* [1976]
K4	117	32.2°S, 295.9°E	8	9.7	82.9	195.5	*Valencio* [1972], *Vilas* [1976]
K3	121	6.4°S, 312.6°E	21	1.9	83.6	261.0	*Schult and Guerreiro* [1979]
K2	126	32.2°S, 295.7°E	10	11.9	66.5	200.7	*Mendia* [1978]
K1	131	29.0°S, 310.0°E	54	3.4	83.8	274.6	*Bellieni et al.* [1983]
J4	164	6.4°S, 312.6°E	15	6.9	85.3	82.5	*Schult and Guerreiro* [1979]
J3	170	48.2°S, 291.2°E	15	8.2	85.7	202.4	*Vilas* [1974]
J2	200	9.0°N, 291.7°E	5	12.1	70.0	121.4	*MacDonald and Opdyke* [1974]
J1	203	7.5°N, 296.8°E	5	4.2	66.8	65.9	*MacDonald and Opdyke* [1974]
	J-K	9.0°N, 293.0°E	9	5.6	86.5	151.1	this study

Ages are K-Ar dates (mega annum) converted to the new decay constant. N is the number of independent estimates of the pole position and A_{95} is the 95% circle of confidence of the site poles (north latitude, east longitude).

motion as inferred from hot spot tracks [e.g., *Morgan*, 1983] which suggest that the South American plate moved northward since the Early Jurassic (Figure 12). For a reference site in the Mérida Andes (9°N, 293°E) the latitudinal change since 170 Ma amounts to 24°, and for a centrally located site (10°S, 300°E) the value is 20°. This large discrepancy exceeds the error estimates of both approaches and suggests that the assumption of the fixedness of hotspots may need to be reevaluated.

Acknowledgments. This work was initiated by Victor Graterol of the Universidad Simón Bolívar, Caracas. His encouragement and interest in this project is greatly appreciated. We thank our colleagues at the University of Texas at Austin for their contributions: Earle McBride and Randy Farr for the petrographic and geochemical analyses of samples from the Quevedo and La Luna Formations; Dietmar Müller for calculating the absolute plate motion for South America; Amos Salvador for sharing his knowledge of Venezuelan geology. We appreciate the discussions with researchers from INTEVEP, LAGOVEN, and CORPOVEN, Venezuela. The paper benefitted from the thoughtful reviews by Myrl Beck and Bill MacDonald. This project was jointly sponsored by CONICIT, Venezuela, and the international program of NSF (grant INT-8608620).

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(Received April 1, 1990;
revised August 31, 1990;
accepted Septenber 10, 1990.)

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